UNIT 7 · TOPIC 10

Alternate Fuels for IC Engines

Syllabus topic: Alternate fuels for IC engines

Motivation (Unit 4, Topic 1b): reduce fossil diesel/petrol use, cost and emissions.

Fuel	Engine	Key points
ETHANOL	SI	E20 blend; ED95/E85 in modified engines; higher octane, lower CV — needs recalibration
BIODIESEL	CI	B5–B20; higher cetane, good lubricity; but higher NOx and cold-flow issues
BIOGAS/CBG and CNG (methane)	SI or dual-fuel	High octane, clean; needs high-pressure storage
PRODUCER GAS	Dual-fuel	From gasifiers (Topic 6)
LPG	SI	Gaseous fuel
HYDROGEN	H_2-ICE, fuel cells	Very high CV; storage challenge
Battery-ELECTRIC	e-tractors	No combustion

Blending/adaptation issues

- Material compatibility (elastomers, metals).
- Energy-density (CV) drop — needs recalibration.
- Cold-start and storage handling.

Ethanol policy (as given in the source notes, July 2026)

- India reached 20% ethanol blending (E20) in November 2025.
- E20 became MANDATORY across all States/UTs from **1 April 2026** (feedstocks: sugarcane, maize, rice/grains) — cuts crude imports. Details: Unit 4, Topic 1b.

Practice MCQs

Q126. Ethanol as an SI-engine fuel has, relative to petrol:

(a) Higher octane, lower calorific value (b) Lower octane
(c) Higher CV (d) No difference

Q127. Biodiesel relative to diesel typically has:

(a) Lower cetane (b) Higher cetane and good lubricity
(c) No cetane (d) Higher sulphur

Q128. A challenge common to alternative fuels is:

(a) Material compatibility and calorific-value adjustment (b) Too much power
(c) No storage need (d) Zero cost

Answers

Q126 — (a) Higher octane, lower calorific value

Q127 — (b) Higher cetane and good lubricity

Q128 — (a) Material compatibility and calorific-value adjustment

Sources: Rai — Non-Conventional Energy Sources; Mittal — Energy in Agriculture; MNRE Biogas/SATAT programmes & PIB (as given in the source notes, July 2026); IS 16087:2016 (CBG specification); standard bioenergy texts; NPTEL bioenergy & energy-management lectures.